

Name: _____

Date: _____

TRANSFORMATIONS, (ENLARGEMENT)

LO: I can enlarge a shape by a given scale factor from a centre of enlargement.

Enlargement

An **enlargement** changes the size of a shape by a **scale factor** from a fixed point called the centre of enlargement.

Key fact: Scale factor > 1 makes the shape bigger. Scale factor between 0 and 1 makes it smaller. Negative scale factor also inverts the image.

Quick examples

●	SF 2 from centre: all distances from centre doubled	shape is twice as big
●	SF $\frac{1}{2}$: image is half the size	reduction
●	SF -1: image inverted through the centre	rotated 180°
●	Enlargement changes angles	angles stay the same

Key idea

Multiply the distance from the centre to each vertex by the scale factor to find the image vertices.

image distance =

SF $\times$ object distance from centre

Measure distances from the centre of enlargement

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – scale factor 2 from origin

Enlarge triangle (1,1), (3,1), (1,3) by SF 2 from (0,0).

Multiply each coordinate by 2
(1,1) (2,2), (3,1) (6,2), (1,3) (2,6)
Image is twice the size

(2,2), (6,2), (2,6)

From the origin, just multiply all coordinates by the SF.

Example 2 – scale factor 3 from a point

Enlarge point (2,1) by SF 3 from centre (1,0).

Vector from centre to point: (2-1, 1-0) = (1, 1)
Multiply by 3: (3, 3)
Image = centre + (3,3) = (4, 3)

(4, 3)

Find the vector from centre to point, multiply by SF, add back to centre.

Example 3 – fractional scale factor

Enlarge (6,4) by SF $\frac{1}{2}$ from centre (0,0).

Multiply each coordinate by $\frac{1}{2}$
(6 $\times$ $\frac{1}{2}$, 4 $\times$ $\frac{1}{2}$)
= (3, 2)

(3, 2)

Fractional scale factors reduce the shape.

Example 4 – negative scale factor

Enlarge (3,1) by SF -2 from centre (1,1).

Vector from centre: (3-1, 1-1) = (2, 0)
Multiply by -2: (-4, 0)
Image = (1,1) + (-4,0) = (-3, 1)

(-3, 1)

Negative SF puts the image on the opposite side of the centre.

YOUR TURN – PRACTICE (deliberate practice)

1. SF 2 from origin

Enlarge each point by SF 2 from (0,0).

- a) (1, 2)
- b) (3, 1)
- c) (2, 4)
- d) (0, 3)

2. SF 3 from origin

Enlarge each point by SF 3 from (0,0).

- a) (1, 1)
- b) (2, 0)
- c) (1, 3)
- d) (4, 2)

3. From a non-origin centre

Enlarge by SF 2 from the given centre.

- a) Point (3,2), centre (1,1)
- b) Point (4,3), centre (2,1)
- c) Point (5,1), centre (3,1)
- d) Point (1,4), centre (0,2)

4. Fractional scale factors

Enlarge by the given SF from (0,0).

- a) (4, 6) SF $\frac{1}{2}$
- b) (9, 3) SF $\frac{1}{3}$
- c) (8, 4) SF $\frac{1}{4}$
- d) (10, 6) SF $\frac{1}{2}$

5. Mixed enlargement

Perform each enlargement.

a) (2,3) SF 2 from (0,0)

b) (4,1) SF 3 from (1,0)

c) (6,8) SF $\frac{1}{2}$ from (0,0)

d) (3,2) SF -1 from (2,2)

e) (5,1) SF -2 from (3,1)

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) An enlargement with SF 2 doubles all lengths ☐

Correct answer: _____

Reason: _____

b) Enlargement with SF 2 doubles the area ☐

Correct answer: _____

Reason: _____

c) Angles remain the same after enlargement ☐

Correct answer: _____

Reason: _____

d) A negative scale factor puts the image on the same side ☐

Correct answer: _____

Reason: _____

e) SF between 0 and 1 makes the shape smaller ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Triangle ABC has vertices A(1,1), B(3,1), C(1,4). It is enlarged by **SF 2 from (0,0)**. Find the area of both triangles and verify that image area = $SF^2 \times$ original area.

Expression: _____

Simplified: _____

b) A shape is enlarged by **SF -2 from centre (3,2)**. The image of point P is at **(-1, 6)**. Find the original position of P.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

